Arman Gheysari

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Research Statement

My research sits at the intersection of distributed systems safety and modern machine learning, organized around three pillars. (1) *Byzantine fault tolerance meets Byzantine-robust ML* — my MSc work and two patents pushed PBFT consensus into System-on-Chip designs, where each processing element pairs with a consensus module that validates committed instructions and reconciles execution results via per-core blockchains and Merkle roots. The same threat model now appears in federated and distributed ML, where malicious workers can poison gradients during training; the defense family (Krum, Bulyan, trimmed mean) mirrors the Byzantine generals problem I worked on in hardware. (2) *Multi-objective optimization for ML systems* — I used genetic algorithms to tune PoW-based blockchain parameters for security-vs-performance tradeoffs (Elsevier 2025), and NSGA-II for multi-objective optimization of Network-on-Chip reliability and latency under buffer size constraints (Springer 2025). The same constrained, non-differentiable, multi-objective search applies directly to ML hyperparameter optimization and hardware-aware distributed training topology design. (3) *Safe offline-to-online reinforcement learning* — my next direction is making offline-trained policies safe to deploy online under hard reliability constraints and adversarial fault patterns, combining the formal verification mindset from BFT work with modern deep RL.

Education

MSc **Amirkabir University of Technology (Tehran Polytechnic)**, Computer Engineering — Computer Architecture | Tehran, Iran | Nov 2020 – Aug 2023

- GPA: 16.43 / 20 · Government scholarship
- Supervisor: Prof. Hamid Reza Zarandi
- Thesis: Byzantine fault-tolerant System-on-Chip design — PBFT consensus modules paired with each processing element, per-core blockchains, and Merkle-tree root verification
- Research focus: Fault tolerance, distributed systems, blockchain, heuristic algorithms, reinforcement learning

BSc **Urmia University of Technology**, Computer Engineering — Information Technology | Urmia, Iran | Oct 2014 – Sept 2019

- GPA: 15.91 / 20 · Government scholarship
- Bachelor's project: Four security scanner plugins (NFS, SMB, FTP, IoT/CCTV) for OWASP-Nettacker, developed in Python

Publications

Security-aware optimization of PoW-based blockchain performance using a genetic algorithm approach | Oct 2025

Configures consensus algorithm and network parameters (block size, block interval, propagation mechanism) using a Genetic Algorithm to enhance performance while preserving resistance to selfish-mining and Eclipse attacks. Simulation-based evaluation iteratively refines configurations to minimize median block propagation time and stale block rate without opening new attack surfaces.

Arman Gheysari, Hamid R. Zarandi

10.1016/j.suscom.2025.101232 (Sustainable Computing: Informatics and Systems (Elsevier))

Simultaneous optimization of network-on-chip to improve reliability and reduce average packet latency considering buffer size constraints Aug 2025

Multi-objective NSGA-II framework for mesh-based NoCs with 5-port routers and XY routing, using buffer size as the main optimization parameter. Validated with Garnet 2.0 on 3×3 to 8×8 networks under uniform, hotspot, and real traffic workloads. On a 4×4 mesh (uniform traffic): 18.0001 cycles average latency and failure rate reduced to 4213 FIT.

Hesam Abdolhosseini, Hamid R. Zarandi, *Arman Gheysari*

www.springerprofessional.de/en/simultaneous-optimization-of-network-on-chip-to-improve-reliabil/51370836 (The Journal of Supercomputing, Vol. 81, No. 13 (Springer))

Optimizing Proof-of-Work Blockchain Network Communications Using Genetic Algorithm Apr 2025

GA-based topology optimization for PoW blockchain networks that systematically refines node placement and interconnections to minimize communication overhead during transaction propagation and block dissemination. Achieves a 20.2% reduction in total data exchanged without compromising core consensus functionality.

Arman Gheysari, Murtadha Almahdawi, Hamid R. Zarandi

ieeexplore.ieee.org/iel8/10967264/10967397/10967452.pdf (IEEE Conference Proceedings)

Patents

Method for Mitigation and Detection of Crash and Byzantine Faults in System-on-Chips Using Blockchain & Consensus Algorithms Patent No. 140250140003006003 Issued Nov 2023

- Fault detection and mitigation system for safety-critical SoCs, incorporating PBFT consensus modules and per-core blockchain storage to ensure correctness and tamper-proof execution
- Merkle tree root verification, a novel arbitration circuit, and a custom algorithm isolate faulty processing units
- Improved system reliability and energy efficiency by removing compromised cores from computation cycles

Fault-Tolerant System-on-Chip Patent No. PCT/IB2024/061908 Filed Nov 2024

- Byzantine fault-tolerant SoC architecture for many-core processors; each processing element (PE) is paired with a consensus module validating SOC controller instructions via PBFT
- Execution results stored in blockchain logs; Merkle tree roots sent to an arbiter module that identifies faulty PEs
- Ensures high system dependability by dynamically purging and reconfiguring around faulty units

In-Progress Research

Safe Offline-to-Online Reinforcement Learning for Fault-Tolerant Distributed Systems. The goal is making offline-trained policies safe to deploy online under hard reliability constraints and adversarial fault patterns. The approach combines uncertainty estimation (so the agent knows when it is in unfamiliar territory) with the structural safety properties already in the PBFT layer underneath it (so that even a wrong RL decision cannot violate consensus safety, only degrade performance). Target: zero unsafe actions during online deployment. The current artifact stack is CQL for offline pretraining, PPO-Lagrangian for constrained online fine-tuning, and a custom PBFT simulator as the environment.

Open Source Projects

rl-for-consensus — Python · Stable Baselines3 · custom PBFT simulator

- A PPO agent that learns to tune PBFT committee size and view-change timeout under Byzantine faults
- Simulator implements the three-phase pre-prepare / prepare / commit protocol from Castro and Liskov (OSDI 1999) with view changes and three attack patterns (collusion, equivocation, timing)
- Result: PPO reaches 0.90 commit rate versus 0.81 for the best static baseline; on the advanced env with attacks enabled, PPO reaches 0.85 versus 0.80

byzantine-robust-fl — Python · PyTorch · NumPy

- Reference implementations of Krum, Bulyan, trimmed mean, median, and geometric median aggregation rules for federated learning
- Four Byzantine attack patterns: sign-flip, label-flip, additive noise, and honest baseline
- Result: On MNIST with 20% Byzantine workers, mean aggregation collapses to 11% accuracy under sign-flip while Krum holds 77%, replicating the headline result from Blanchard et al. (NeurIPS 2017)

safe-rl-playground — Python · Stable Baselines3 · Gymnasium

- PPO, PPO-Lagrangian (constrained), and CQL (offline) on safety-augmented Gymnasium environments
- SafeCartPole adds a soft cost on pole angle exceeding a threshold (0.05 rad); SafeMountainCar adds a cost on velocity exceeding a threshold
- Studies the trade-off between reward and safety: when they align, PPO suffices; when they conflict, PPO-Lagrangian becomes necessary; CQL trains offline on a random-policy dataset

Talks & Presentations

- IEEE Conference (Apr 2025) — *Optimizing Proof-of-Work Blockchain Network Communications Using Genetic Algorithm*. Conference presentation.
- *Upcoming:* In preparation — submissions on safe offline-to-online RL for fault-tolerant distributed systems.

Teaching Experience

Amirkabir University of Technology, Teaching Assistant — Fault Tolerance — Tehran, Iran — Feb 2025 – June 2025

- Course presented by Prof. Hamid Reza Zarandi
- Grading, leading recitation sessions, and supporting MSc-level fault-tolerance coursework (BFT, consensus, dependability evaluation)

Amirkabir University of Technology, Teaching Assistant — Testing & Testable Design — Tehran, Iran — Feb 2023 – June 2023

- Course presented by Prof. Hamid Reza Zarandi
- Course coverage: fault models, test generation, ATPG, scan chains, BIST, and design-for-testability

Amirkabir University of Technology, Lab Instructor — Operating Systems — Tehran, Iran — Feb 2021 – June 2021

- Designed and delivered hands-on labs in C/C++, Python, and Ubuntu Linux covering processes, threads, synchronization, and file systems

Urmia University of Technology, Instructor — Robotic Training Course — Urmia, Iran — Sept 2018 – Dec 2018

- Designed and led an undergraduate robotics course built on Python, Linux, and Raspberry Pi

Professional Experience

Amirkabir University of Technology — Computing & Information Center (CIC), Security & Data Analyst — Tehran, Iran — July 2024 – present

- Security monitoring, threat analysis, and data-driven reporting for the central IT infrastructure of the university
- Built detection pipelines and dashboards on university network telemetry; collaborate with the IT security team on incident response

Amirkabir University of Technology, Research Assistant — Tehran, Iran — Sept 2020 – present

- Fault-tolerant SoC design, blockchain-based consensus for hardware, and bio-inspired optimization of distributed-system parameters under Prof. Hamid Reza Zarandi
- Worked on PBFT consensus modules, per-core blockchains, Merkle root verification, and arbitration circuits for many-core processors — outcomes captured in two patents and three publications

Central Telecommunications Company of West Azerbaijan, Network Operations Intern — Urmia, Iran — June 2017 – Sept 2017

- Network operations and infrastructure work at the regional telecom provider; hands-on with transmission equipment and field diagnostics

Research & Engineering Projects

- Implementing a Blockchain Using PoW Consensus Algorithm — Python blockchain built from scratch with Proof-of-Work consensus. *Aug 2023*
- Implementing a Blockchain Using PBFT Voting-Based Consensus Algorithm — Practical Byzantine Fault Tolerance voting-based consensus. *Dec 2022*
- 64-bit UNIX-like Operating System (MIT 6.828 JOS) — Built from scratch on the MIT 6.828 JOS framework. *Oct 2021*
- Parking System on FPGA via HW/SW Co-design — Xilinx Vivado-based parking system combining hardware and software components. *Sep 2021*
- Python Compiled-Code Compiler — Lexer, parser, and code generation in Python. *Jan 2021*
- Event-Driven Compiler in Python — Event-based code generation pipeline. *Jan 2021*
- Four OWASP-Nettacker Plugins — Security scanner plugins (NFS, SMB, FTP, IoT/CCTV) for OWASP-Nettacker, in Python. Bachelor's project. *Feb 2019*
- Routing Algorithm with Genetic Algorithm — AI course project on GA-based routing. *Dec 2018*
- 3D-Map Generator with Kinect & ROS — 3D-map generator using Microsoft Kinect sensor and Robot Operating System. *Nov 2018*
- Python Framework for Motors, Servo Motors & Actuators — Reusable Python framework for actuator control. *Nov 2018*
- Python Scripts for Reading Data from Various Sensors — Sensor I/O toolkit for robotics. *Nov 2018*
- Object Detection with YOLO & OpenCV — Real-time object detection with OpenCV, YOLO algorithm, and Python. *Oct 2018*
- Path Planning in ROS for Rescue Robot — Path planning algorithm in ROS for a rescue robot navigation system. *Oct 2018*
- C Compiler in C++ — Front-end and codegen for a C-subset compiler, written in C++. *Jan 2016*

Honors & Awards

- Ranked 3rd among 13 peer M.Sc. students in Computer Engineering (Computer Architecture), Amirkabir University of Technology
- Ranked 190th among 20,000–25,000 participants in the Iranian nation-wide M.Sc. entrance exam for Computer Engineering
- Head of Python Programming Team, Urmia University of Technology Robotics Team

Certifications

- Fundamentals of Reinforcement Learning — University of Alberta (Coursera). Verify Certificate
- Sample-based Learning Methods — University of Alberta (Coursera). Verify Certificate
- Prediction and Control with Function Approximation — University of Alberta (Coursera). Verify Certificate
- Modern Robotics: Mechanics, Planning, and Control Specialization — Coursera
- LPIC-1 Certified Linux Administrator — Linux Professional Institute. Verify Certificate · *Dec 2016*
- CCNA Routing & Switching — Cando Training Center · *Apr 2016*
- Network+ Certified — Sharif University of Technology (LAITEC) · *Feb 2016*
- ISMS Auditor (ISO 19011:2011 / ISO/IEC 27001:2013) — Batis Academy · *Jan 2017*

Skills

Machine Learning & RL: PyTorch, Stable Baselines3, Gymnasium / OpenAI Gym, D4RL, Flower (FL), TensorFlow, Keras, NumPy, Pandas, Scikit-Learn, Optuna, Matplotlib, Seaborn

Optimization & Search: NSGA-II (DEAP / pymoo), Genetic Algorithms, Multi-objective Optimization, Constrained Optimization, Pareto Front Analysis

Distributed Systems & Security: PBFT / Byzantine Fault Tolerance, Byzantine-Robust Aggregation (Krum, Bulyan, trimmed mean), Consensus Protocols, Blockchain (PoW), Merkle Trees, Hardware Security

Programming: Python, C / C++, HDL / SystemC, Bash, Perl, Network-Datalog (NDLog), Git / GitHub, Linux, Docker, ROS (Robotic Operating System)

Simulators & Hardware: GEM5, NS-3, BFTSim, RapidNet, Noxim, Modelsim, Cisco Packet Tracer, Wireshark, Burp Suite, Xilinx Vivado

Networking & Sysadmin: CCNA Routing & Switching, LPIC-1 / LPIC-2, Advanced Penetration Testing with Kali Linux, ISMS (ISO 19011:2011 / ISO/IEC 27001:2013)

Notable Courses

- Advanced Operating System — 19.32 / 20
- Advanced Computer Architecture — 19.10 / 20
- Testing & Testable Design — 18.21 / 20
- Basics of AI and Applied Artificial Intelligence — 17.50 / 20
- Fault Tolerance — 16.35 / 20

Languages

Persian: Native

English: Second Language — professional working proficiency